

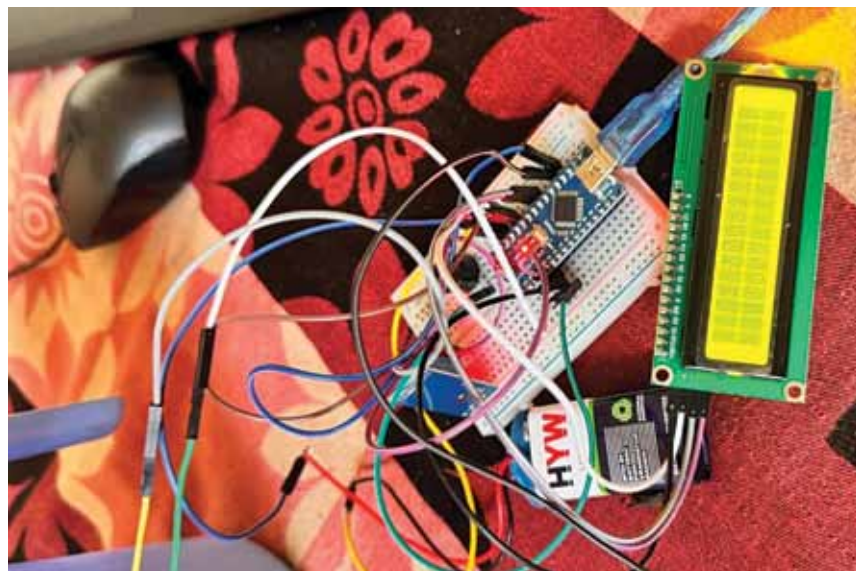


Fig. 4: Author's prototype after connection

a stable power supply, and the Nano itself is connected to the computer via USB.

Once the hardware setup is complete, the circuit is powered, and the LCD initialisation is observed. For testing, waste is

gradually added to the EcoBin. When the ultrasonic sensor detects that the bin is full, the buzzer sounds. The MQ135 sensor triggers an alarm if smoke or harmful gases are present and also monitors air quality. The DHT11 sensor provides

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temperature and humidity data, all of which are displayed in real time on the LCD. To verify proper functioning, sensor outputs are cross-checked on the Arduino serial monitor. This ensures that the Smart EcoBin operates reliably and delivers timely alerts for waste level, air quality, and environmental conditions. **EFY**



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